

WINNIPEG THE FORKS, MB, Canada

WMO: 715790

Lat: 49.8885N

Lon: 97.1296W

Elev: 230

StdP: 98.59

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.1	(c) -26.1	(d) -32.4	(e) 0.2	(f) -27.9	(g) -30.3	(h) 0.2	(i) -25.9	(j) 7.9	(k) -10.6	(l) 6.9	(m) -12.6	(n) 2.0	(o) 210	(p) 0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 30.8	(d) 20.7	(e) 29.2	(f) 19.8	(g) 27.7	(h) 18.9	(i) 22.7	(j) 28.2	(k) 21.5	(l) 27.2	(m) 20.3	(n) 25.6	(o) 2.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	16.0	26.4	19.5	14.6	24.9	18.3	13.5	23.7	68.6	28.1	63.8	27.1	59.6	25.6	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.7	(b) 5.8	(c) 5.1	(e) -31.0	(f) 34.0	(g) 2.4	(h) 1.5	(i) -32.7	(j) 35.1	(k) -34.1	(l) 36.0	(m) -35.4	(n) 36.8	(o) -37.1	(p) 37.9		
(a) 6.7	(b) 5.8	(c) 5.1	(e) -31.2	(f) 24.9	(g) 2.3	(h) 1.5	(i) -32.8	(j) 26.0	(k) -34.2	(l) 26.9	(m) -35.5	(n) 27.8	(o) -37.2	(p) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-13.9	-12.7	-4.2	5.2	12.3	18.2	21.5	20.2	14.9	6.7	-2.1	-11.0		
	DBStd	13.81	7.72	7.04	7.41	6.13	5.12	3.69	3.12	3.41	4.55	5.02	6.26	7.17		
	HDD10.0	3176	741	636	443	166	36	1	0	0	10	128	363	652		
	HDD18.3	5238	999	869	700	396	195	47	9	20	121	360	613	911		
	CDD10.0	1234	0	0	2	21	107	248	357	315	158	27	1	0		
	CDD18.3	254	0	0	0	0	8	44	107	77	18	1	0	0		
	CDH23.3	2043	0	0	0	3	102	345	836	604	144	9	0	0		
	CDH26.7	552	0	0	0	0	22	77	238	183	32	1	0	0		
(14) Wind	WSAvg	2.3	2.4	2.3	2.4	2.6	2.5	2.1	2.0	2.0	2.1	2.3	2.3	2.3		
(15) Precipitation	PrecAvg	523	19	15	25	36	61	83	77	79	50	35	23	20		
	PrecMax	777	52	49	68	99	181	232	193	222	151	129	62	42		
	PrecMin	237	3	0	1	0	8	6	20	6	10	2	1	3		
	PrecStd	111	12	12	15	26	45	47	46	50	30	28	16	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	5.0	17.0	23.3	29.3	31.1	33.0	33.1	30.0	24.5	14.7	4.3		
		MCWB	2.1	2.7	10.1	11.9	16.2	20.6	22.8	21.6	19.7	15.4	9.5	1.7		
	2%	DB	1.7	2.7	11.0	19.9	26.1	28.9	30.8	30.7	27.1	20.0	10.7	1.9		
		MCWB	0.3	0.8	6.9	10.7	14.6	18.8	21.3	20.6	18.5	12.8	6.7	0.4		
	5%	DB	-0.4	0.5	7.6	17.5	23.5	27.0	29.2	28.7	24.7	16.8	8.5	0.2		
		MCWB	-1.5	-0.8	4.4	9.4	13.6	17.7	20.4	19.8	17.2	10.9	5.6	-0.8		
	10%	DB	-2.8	-2.4	4.8	14.9	20.7	25.2	27.8	26.8	22.1	14.2	5.9	-1.4		
		MCWB	-3.8	-3.6	2.5	8.1	12.4	17.1	19.5	18.7	16.0	9.8	3.5	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	3.2	11.7	13.7	18.8	23.1	24.9	24.4	21.9	16.1	10.1	2.0		
		MCDB	3.9	4.9	15.2	20.8	25.1	28.0	31.0	29.5	27.5	22.8	13.2	3.8		
	2%	WB	0.5	1.0	7.1	11.4	16.4	20.9	23.2	22.1	19.9	13.8	7.4	0.6		
		MCDB	1.7	2.2	11.0	18.3	21.2	26.2	28.6	28.4	24.4	18.9	10.3	1.8		
	5%	WB	-1.5	-1.0	4.5	9.9	15.1	19.3	22.0	21.0	18.3	11.7	5.4	-0.8		
		MCDB	-0.4	0.4	7.2	16.0	20.5	24.6	27.5	26.9	22.8	15.9	8.1	0.1		
	10%	WB	-3.8	-3.5	2.7	8.5	13.7	18.0	20.6	19.8	16.9	9.9	3.6	-2.3		
		MCDB	-2.8	-2.4	4.8	14.5	19.5	23.0	25.9	24.9	21.2	14.0	5.8	-1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.1	8.5	10.0	10.4	9.5	9.5	9.8	9.5	7.5	6.5	6.6		
		MCDBR	8.6	8.4	10.4	14.0	14.3	12.1	11.2	12.5	13.3	12.0	9.8	7.7		
		MCWBR	7.7	7.2	7.3	7.5	6.5	5.5	5.3	5.6	6.7	6.6	6.9	6.8		
		MCDBR	8.6	8.1	9.8	12.7	10.9	10.1	10.5	11.0	10.9	9.3	7.5	7.5		
	5% WB	MCWBR	7.7	7.0	7.1	7.3	6.1	5.8	5.9	5.7	6.7	6.6	6.7	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.302	0.308	0.357	0.373	0.386	0.393	0.381	0.348	0.306	0.275	0.252		
	taud		2.245	2.137	2.297	2.330	2.352	2.340	2.338	2.368	2.455	2.537	2.464	2.327		
	Ebn at Noon		774	824	899	887	885	872	860	853	850	836	781	752		
	Edh at Noon		71	103	105	115	118	120	119	110	91	70	58	57		
(44) All-Sky Solar Radiation	RadAvg		1.35	2.57	3.84	5.01	5.45	5.83	6.15	5.06	3.65	2.16	1.19	0.96		
	RadStd		0.15	0.19	0.32	0.38	0.45	0.45	0.26	0.34	0.28	0.29	0.17	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-1.06	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.79	N/A	N/A	N/A	N/A

Nomenclature: See separate page